Hard submatrices for non-negative rank

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Editorial note. Faithful transcription from official CCC 2024 CC BY 4.0 PDF; original notation, indexing and dimensions retained. Author TeX was not located. No author proof is replaced or extended; standard cited prior results are retained as prerequisites, and bibliography entries keep their original numbers.

Allowed standard prerequisites (editorial).

- Slack-matrix/extension-complexity sandwich bound. rk_+(S)-1 <= xc(P0,P1) <= rk_+(S). Braun et al. [2], source Lemma 14; exact statement retained
- Single existential-block quantifier elimination bound. Basu, Pollack and Roy [1], Theorem 14.16; exact used statement retained in appendix

3 A combinatorial lemma (author source pp. 13:3–13:4)

Both Theorems 1 and 2 rely on a simple combinatorial lemma.

Lemma 3. *Let $\mathcal{A} \subseteq 2^{[n]}$ be a family of subsets of $[n]$. Assume that $1 \leq k \leq n$ is such that every k-element subset of $[n]$ is contained in some $A \in \mathcal{A}$. Then there exists a subfamily $\mathcal{A}' \subseteq \mathcal{A}$ of size $|\mathcal{A}'| \leq O(|\mathcal{A}|^{1/k} \log(n/k))$ with $\bigcup \mathcal{A}' = [n]$. In particular, if $|\mathcal{A}| \leq 2^k$ then $|\mathcal{A}'| \leq O(\log n)$.*

Proof. Assume that $|\mathcal{A}| \leq a^k$. Let t be the size of a largest set in $\mathcal{A}$. Then we have

$$\binom{n}{k} \leq a^k \binom{t}{k}.$$

Hence $t \geq \frac{n}{ea}$, using the estimates $(\frac{n}{k})^k \leq \binom{n}{k}$, $\binom{t}{k} \leq (\frac{et}{k})^k$. Take some $A_0 \in \mathcal{A}$ of size t. Let

$$\mathcal{A}_1 := \{A \setminus A_0 : A \in \mathcal{A}\}.$$

Then every subset of $U_1 := [n] \setminus A_0$ of size at most k is contained in some member of $\mathcal{A}_1$. The size of U_1 is at most $n(1 - \frac{1}{ea})$. Similarly, take a largest set A_1 from $\mathcal{A}_1$ and obtain a new family $\mathcal{A}_2 \subseteq 2^{U_2}$ on $U_2 := U_1 \setminus A_1$. After s steps, the size of U_s is at most $n(1 - \frac{1}{ea})^s$ and after $s \leq O(a \log(n/k))$ steps we have $|U_s| \leq k$. This guarantees that the largest set in $\mathcal{A}_s$ is U_s itself and $[n] = \bigcup_{i=0}^{s} A_s$. By construction, every A_i is contained in some element of the original family $\mathcal{A}$. □

6.1 Extended formulations and separation complexity (author source p. 13:8)

Let us first make a short detour into extended formulations of convex polyhedra. A polyhedron $P \subseteq \mathbb{R}^r$ is a (possibly unbounded) set defined by a finite number of linear constraints. Following [20,18,2], define the extension complexity of P, $\mathrm{xc}(P)$, as the smallest s such that P is a linear projection of a polyhedron $Q \subseteq \mathbb{R}^m$ where Q can be defined using s inequalities (and any number of equalities). Observe that P with extension complexity s can be expressed in the standard form

$$x \in P \text{ iff } \exists y \in \mathbb{R}^s \; Cx + Dy = b, \; y \geq 0,$$

where $C \in \mathbb{R}^{t \times r}$, $D \in \mathbb{R}^{t \times s}$ and $b \in \mathbb{R}^t$ for some t. Let V be a finite subset of $\mathbb{R}^r$. Given $A \subseteq V$, its separation complexity, $\mathrm{sep}_V(A)$, is the minimum $\mathrm{xc}(P)$ over all polyhedra $P \subseteq \mathbb{R}^r$ with[1]

$$P \cap V = A;$$

such a P is called a separating polyhedron for A. In other words, $\mathrm{sep}_V(A)$ is the smallest s so that we can distinguish points in A from points in $V \setminus A$ by means of a linear program with s inequalities. Moreover, such a program can be rewritten as

$$x \in A \text{ iff } (x \in V \text{ and } \exists y \in \mathbb{R}^s \; Cx + Dy = b, \; y \geq 0).$$

The notion of separation complexity has been studied in [7,8,9] in the case when $V = \{0,1\}^n$ is the Boolean cube. The following theorem is of independent interest and can be seen as an extension of similar results in [7,9]. The proof is a considerable simplification of the previous ones.

Theorem 13 (author source p. 13:8). *Let V be a non-empty finite subset of $\mathbb{R}^r$. Given a parameter $s \geq 1$, let $\mathcal{A}$ be the collection of subsets A of V with $\mathrm{sep}_V(A) \leq s$. Then*

$$|\mathcal{A}| \leq 2^{O(s(r+s)^2 \log |V|)}.$$

A Proof of Theorem 13 (author source pp. 13:11–13:12)

The proof uses known results on quantifier elimination which we first outline. We follow the monograph of Basu, Pollack and Roy [1]. Theorem 13 requires an elimination of only a single block of existential quantifiers, so we focus on this case only. For $b \in \mathbb{R}$, let

$$\mathrm{sgn}(b) := \begin{cases} 1, & b > 0, \\ 0, & b = 0, \\ -1, & b < 0. \end{cases}$$

[1] If no such polyhedron exists, which may happen if V is not convexly independent, we set $\mathrm{sep}_V(A) := \infty$.

Given $b = \langle b_1, \ldots, b_m \rangle \in \mathbb{R}^m$, let $\mathrm{sgn}(b) := \langle \mathrm{sgn}(b_1), \ldots, \mathrm{sgn}(b_m) \rangle \in \{-1, 0, 1\}^m$. Let $F = F(z, y)$ be a sequence of m polynomials $f_1, \ldots, f_m \in \mathbb{R}[z, y]$ in variables $z = \{z_1, \ldots, z_{k_1}\}$ and $y = \{y_1, \ldots, y_{k_2}\}$. Given $a \in \mathbb{R}^{k_1}$, define $\mathrm{SGN}_1(F, a) \subseteq \{-1, 0, 1\}^m$

$$\mathrm{SGN}_1(F, a) := \{\mathrm{sgn}(F(a, b)) : b \in \mathbb{R}^{k_2}\}.$$

Let

$$\mathrm{SGN}(F) := \{\mathrm{SGN}_1(F, a) : a \in \mathbb{R}^{k_1}\}.$$

Theorem 14.16 from [1] provides the following bound on the size of SGN:

Theorem ([1]). *If every polynomial in F has degree at most d then*

$$|\,\mathrm{SGN}(F)| \leq m^{(k_1+1)(k_2+1)} d^{O(k_1)O(k_2)}. \tag{4}$$

We now apply this result to the case of Theorem 13. Let $V, s, \mathcal{A}$ be as in the assumption. Every $A \in \mathcal{A}$ can be described by a linear system with s inequalities. Namely, for every $x \in V$,

$$x \in A \text{ iff } \exists y \in \mathbb{R}^s \; Cx + Dy = b, \; y \geq 0, \tag{5}$$

where $C \in \mathbb{R}^{t \times r}$, $D \in \mathbb{R}^{t \times s}$ and $b \in \mathbb{R}^t$. Since $Cx + Dy = b$ is a system of equations in $r + s$ variables x, y, we can also assume $t = r + s$. Let us view the parameters C, D, b in (5) as variables. Let z be the set of these variables, of size $k_1 = (r + s)(r + s + 1)$. Given $v \in V$, let $F_v(z, y)$ be the sequence of $(r + s)$ polynomials

$$Cv + Dy - b$$

in variables z and $y = \{y_1, \ldots, y_s\}$. Let $F(z, y)$ be the union of $F_v(z, y)$ over all $v \in V$, together with the polynomials $y_1, \ldots, y_s$. Hence F consists of $m = s + |V|(r + s)$ polynomials of degree at most two. $F(z, y)$ is set up so that

$$|\mathcal{A}| \leq |\,\mathrm{SGN}(F)|.$$

To see this, observe that whenever the parameters z are fixed, the set $A \subseteq V$ given by (5) is uniquely determined by $\mathrm{SGN}_1(F(z, y))$. Since every $A \in \mathcal{A}$ is obtained by some fixing of the parameters, we indeed obtain $|\mathcal{A}| \leq |\,\mathrm{SGN}(F(z, y))|$. Finally, we can apply (4) to estimate $|\,\mathrm{SGN}(F)|$ with $m = s + |V|(r + s)$, $k_1 = (r + s)(r + s + 1)$, $k_2 = s$, and $d = 2$. To simplify the expression, we can assume $s + r \leq |V|$; otherwise the upper bound asserted in Theorem 13 exceeds the trivial bound $|\mathcal{A}| \leq 2^{|V|}$. This means that $m \leq 2|V|^2$. If we loosen the bound (4) as $|\,\mathrm{SGN}(F)| \leq (dm)^{O(k_1)O(k_2)}$, we obtain (recall that $s \geq 1$)

$$|\,\mathrm{SGN}(F)| \leq 2^{O(s(s+r)^2 \log |V|)},$$

as required.

Non-negative rank (author source p. 13:2)

Given an $m \times n$ matrix M with real non-negative entries, its non-negative rank, $\mathrm{rk}_+(M)$, is the smallest s such that M can be written as

$$M = LR,$$

where L and R are non-negative matrices of dimensions $m \times s$ and $s \times n$, respectively.

Notation. All logarithms are in base 2 and $[n] := \{1, \ldots, n\}$.

6.2 Submatrices of large non-negative rank (author source pp. 13:8–13:9)

In order to apply Theorem 13, we also need a connection between extension complexity and non-negative rank. This is provided by the notion of slack matrix introduced in [20]. Following [20,2], we now define what it is. Let V be a sequence $v_1, \ldots, v_{m_1}$ of points in $\mathbb{R}^r$ and $L(x)$ a system $\ell_1(x) \geq b_1, \ldots, \ell_{m_2}(x) \geq b_{m_2}$ of inequalities in $\mathbb{R}^r$. The slack matrix with respect to V and $L(x)$ is the $m_2 \times m_1$ matrix S such that

$$S_{i,j} = \ell_i(v_j) - b_i.$$

Let $P_0 := \text{conv}(V)$ be the convex hull of V and $P_1 := \{x \in \mathbb{R}^r : L(x) \text{ holds}\}$. If $P_0 \subseteq P_1$ then S is non-negative. In [2], we can find:

Lemma 14 ([2]). *Let $P_0 \subseteq P_1$ and S be as above. Define $\text{xc}(P_0, P_1)$ as the minimum $\text{xc}(P)$ over all polyhedra with $P_0 \subseteq P \subseteq P_1$. Then*

$$\text{rk}_+ S - 1 \leq \text{xc}(P_0, P_1) \leq \text{rk}_+ S.$$

Subadditivity (author source p.13:4). Observe that[2]

$$\chi_1(M_{A_1 \cup A_2}) \leq \chi_1(M_{A_1}) + \chi_1(M_{A_2}), \tag{1}$$

and so $\chi_1(M_A)$ can be viewed as a subadditive measure on $[n]$ whenever M is fixed.

Theorem 1 (restated; author source p. 13:9). *Let M be an $m \times n$ non-negative real matrix with $n \geq 2$. Then for every $k \leq n$, M contains an $m \times k$ submatrix of k columns with non-negative rank $\Omega(R)$, where $R := \min((\frac{k}{\log n})^{1/3}, \frac{\text{rk}_+(M)}{\log n})$.*

Proof. Let r be the rank of M. We can write $M = LR$ where $L \in \mathbb{R}^{m \times r}$, $R \in \mathbb{R}^{r \times n}$. Let $V \subseteq \mathbb{R}^r$ be the set of columns $v_1, \ldots, v_n$ of R. (Without loss of generality, the columns of M are distinct). Given $A \subseteq [n]$, let M_A be the submatrix obtained by deleting columns outside of A from M. Also let $V_A := \{v_i : i \in A\}$. Then M_A can be interpreted as the slack matrix of the polytope $P_A = \text{conv}(V_A)$ and the polyhedron $Q = \{x \in \mathbb{R}^d : Lx \geq 0\}$. Suppose that for every A of size k, $\text{rk}_+(M_A) \leq s$. Then for every such A, there is a polyhedron Q_A with $V_A \subseteq Q_A \subseteq Q$ with $\text{xc}(Q_A) \leq s$. Let $A^* := V \cap Q_A$. Then Q_A is a separating polyhedron for $A^* \supseteq A$. Let $\mathcal{A}$ be the collection of A^* over all A of size k. Theorem 13 implies

$$|\mathcal{A}| \leq 2^{c \log n (s+r)^3},$$

[2] If A_1, A_2 are disjoint, this is quite obvious. Otherwise consider $A_1, A_2 \setminus A_1$.

where c is an absolute constant. We will assume $r \leq (\frac{k}{2c \log n})^{1/3}$. Otherwise M contains a full rank $r \times r$ submatrix, rk_+ is lower-bounded by rank, and the conclusion of the theorem follows. If $|\mathcal{A}| \geq 2^k$, we obtain $c \log n (s+r)^3 \geq k$ and hence $s \geq \Omega((k/\log n)^{1/3})$ from the assumption on r. Assume $|\mathcal{A}| \leq 2^k$. By Lemma 3, there exists a subfamily $\mathcal{A}' \subseteq \mathcal{A}$ of size $O(\log n)$ which covers $[n]$. This implies (note that (1) holds also for non-negative rank) $\mathrm{rk}_+(M) \leq O(s \log n)$ and $s \geq \Omega(\mathrm{rk}_+(M)/\log n)$. □

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